

NAPA COUNTY AP, CA, USA

WMO: 724955

Lat: 38.210N

Lon: 122.285W

Elev: 4

StdP: 101.27

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 93227

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.3	0.1	-8.0	1.9	10.9	-5.4	2.4	9.6	11.2	13.3	9.7	12.9	0.9	0	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	32.8	18.9	30.2	18.3	28.0	17.8	20.2	30.0	19.3	28.3	18.6	26.5	4.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	11.7	23.5	15.8	11.2	22.6	15.1	10.7	21.7	57.8	30.1	54.8	28.3	52.4	26.6	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.5	7.8	DB	-4.6	37.6	1.3	2.0	-5.5	39.0	-6.3	40.2	-7.1	41.3	-8.0	42.7	(4)
(5)			WB	-5.3	22.0	1.4	0.8	-6.3	22.5	-7.1	23.0	-7.9	23.5	-8.9	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	8.6	9.9	11.5	12.9	15.4	18.1	18.8	19.0	18.4	15.6	11.5	8.5	(6)
(7)		DBStd	4.64	2.50	2.58	2.65	2.79	2.58	2.61	2.11	2.01	2.63	2.59	2.89	2.86	(7)
(8)		HDD10.0	188	57	30	14	6	0	0	0	0	0	0	17	64	(8)
(9)		HDD18.3	1709	302	235	211	164	98	34	17	14	30	92	206	306	(9)
(10)		CDD10.0	1663	14	28	62	92	168	242	274	278	252	174	61	17	(10)
(11)		CDD18.3	141	0	0	1	1	7	27	33	33	32	7	0	0	(11)
(12)		CDH23.3	2236	1	2	19	71	184	421	370	396	514	242	14	1	(12)
(13)		CDH26.7	799	0	0	2	16	65	162	122	133	221	76	1	0	(13)
(14)	Wind	WSAvg	3.4	2.8	2.9	3.1	3.6	4.1	4.4	4.5	4.1	3.3	2.6	2.4	2.8	(14)
(15)	Precipitation	PrecAvg	671	137	116	95	39	18	5	1	3	8	38	94	122	(15)
(16)		PrecMax	1303	350	388	304	144	94	50	27	33	59	263	267	393	(16)
(17)		PrecMin	266	9	5	3	2	0	0	0	0	0	0	4	0	(17)
(18)		PrecStd	220	91	94	73	37	25	10	4	7	13	45	69	91	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.5	22.3	25.7	28.8	32.3	35.1	33.9	34.5	35.7	32.2	25.2	18.3	(19)
(20)			MCWB	10.5	14.5	16.7	17.7	18.6	20.0	20.2	19.4	18.6	16.5	15.4	9.8	(20)
(21)		2%	DB	16.9	19.5	22.6	25.3	28.1	31.1	29.8	30.1	32.5	28.8	22.4	16.6	(21)
(22)			MCWB	11.2	13.1	15.0	16.3	17.3	18.7	19.5	19.2	18.2	15.6	13.9	11.4	(22)
(23)		5%	DB	15.0	17.4	20.0	22.2	24.8	27.8	27.1	27.4	29.1	26.0	19.8	15.2	(23)
(24)			MCWB	11.1	11.9	13.6	14.9	16.2	17.8	18.6	18.5	17.5	14.8	13.0	11.7	(24)
(25)		10%	DB	13.5	15.6	17.8	19.5	22.2	25.2	24.9	25.1	25.9	23.3	17.6	14.0	(25)
(26)			MCWB	10.8	11.6	12.8	13.9	15.4	17.0	17.7	17.8	17.2	14.5	12.5	11.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	15.6	17.8	19.1	20.4	21.2	21.2	20.9	20.6	18.8	17.2	15.0	(27)
(28)			MCDWB	15.8	19.7	24.1	26.9	30.2	32.9	31.8	31.0	30.7	27.1	23.1	16.0	(28)
(29)		2%	WB	12.9	14.3	15.8	17.3	18.1	19.6	19.8	19.7	19.4	17.5	15.6	13.9	(29)
(30)			MCDWB	14.7	17.8	21.2	24.4	26.1	29.2	29.0	28.8	29.1	24.7	19.6	14.9	(30)
(31)		5%	WB	12.1	13.3	14.6	15.6	16.9	18.4	18.8	18.8	18.5	16.6	14.6	12.9	(31)
(32)			MCDWB	13.6	16.1	18.6	21.0	23.6	26.5	26.6	26.7	27.0	23.0	17.7	13.9	(32)
(33)		10%	WB	11.3	12.3	13.4	14.3	15.7	17.3	17.9	17.9	17.6	15.6	13.6	11.7	(33)
(34)			MCDWB	13.0	14.7	16.7	18.5	21.6	24.5	24.6	24.8	25.2	21.3	16.5	13.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.1	11.1	12.1	13.0	13.4	14.0	12.9	13.3	15.5	15.3	12.6	10.4	(35)
(36)			MCDBR	13.2	15.2	16.6	18.2	19.0	19.4	17.3	18.2	21.7	20.8	16.5	12.1	(36)
(37)			MCWBR	8.5	9.4	10.1	10.5	9.8	8.9	8.1	8.4	9.3	9.7	9.2	7.7	(37)
(38)		5% WB	MCDWB	8.9	11.9	14.3	15.9	16.4	17.0	15.9	16.4	18.4	16.3	12.2	7.1	(38)
(39)			MCWBR	6.5	7.9	9.3	9.8	9.2	8.2	7.6	7.8	8.5	8.5	7.8	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.321	0.334	0.356	0.360	0.357	0.361	0.345	0.343	0.321	0.318	0.321	(40)	
(41)		taud	2.548	2.548	2.501	2.411	2.430	2.444	2.429	2.468	2.477	2.576	2.560	2.546	(41)	
(42)		Ebn at Noon	865	913	929	922	920	919	912	921	907	900	861	836	(42)	
(43)		Edh at Noon	76	86	99	114	114	113	114	107	100	82	74	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.30	3.26	4.62	6.09	7.21	7.97	7.76	6.91	5.73	4.11	2.75	2.01	(44)	
(45)		RadStd	0.37	0.42	0.46	0.51	0.44	0.34	0.24	0.23	0.20	0.28	0.19	0.32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.41	N/A	-1.67	N/A	N/A	N/A	N/A	N/A	+141	N/A	(46)	
(47)	Regional (47 neighbors)	+0.37	N/A	-1.15	N/A	N/A	N/A	N/A	-115	+133	N/A	(47)	

Nomenclature: See separate page